Question #1 of 49

Question ID: 1473319

An analyst values a private company using a price multiple based on recent sales of comparable assets. This approach to private company valuation is *best* described as the:

A) income approach

B) market approach

C) asset-based approach

Explanation

Under the market approach, a firm is valued using price multiples based on recent sales of comparable assets. Under the income approach, a firm is valued according to the present value of its expected future income. Under the asset-based approach, the value of a firm is calculated as the firm's assets minus its liabilities.

(Module 24.1, LOS 24.d)

Question #2 of 49

Question ID: 1473327

Which of the following *best* describes how debt is incorporated into the estimation of the discount rate for private company valuations, relative to that for public firms? In general, the cost of debt:

A) is higher for private firms and debt capacity is lower for private firms.

B) and debt capacity is the same for both private and public firms.

C) is higher for private firms and debt capacity is the same for both private and public firms.

Explanation

A private firm may not be able to obtain as much debt financing as a public firm. The small size of private firms may result in higher operating risk and a higher cost of debt.

(Module 24.2, LOS 24.e)

Question #3 of 49

Question ID: 1473330

Which of the following *best* describes the build-up method used for the estimation of the discount rate in private company valuations?

A) An industry risk premium is not included because it is captured in the equity risk premium.

B) It is useful when there are no comparable public firms.

C) Because it is not used in the calculation, beta is assumed to be zero.

Explanation

If it is not possible to find comparable public firms with which to estimate beta by, the build-up method can be used for a private firm. It is similar to the expanded CAPM except that beta is not used. Implicitly, beta is assumed to be one. Both industry risk premiums and equity risk premiums are used. The risk-free rate, the equity risk premium, the small stock premium, a company-specific risk premium, and an industry risk premium are added together in the build-up method.

(Module 24.2, LOS 24.f)

Question #4 of 49

Question ID: 1473316

A private business is being valued for the purpose of determining the appropriate level of performance-based managerial compensation. This private company valuation would be *best* described as a:

A) Compliance-related valuation

B) Litigation-related valuation

C) Transaction-related valuation

Explanation

Transaction-related valuations may be performed for reasons related to venture capital financing, an IPO, a sale of the firm, bankruptcy, or performance-based managerial compensation. Compliance-related valuations are performed for financial reporting and tax purposes. Litigation-related valuations may be required for shareholder suits, damage claims, lost profits, or divorces.

(Module 24.1, LOS 24.b)

Question #5 of 49

Question ID: 1473329

Which of the following *best* describes projection risk in the estimation of the discount rate for private company valuations?

A) If the availability of information from private firms is poor, the uncertainty of projected cash flows may increase.

B) Projection risk results in higher discount rates.

C) Management will always be overly optimistic to increase the acquisition price.

Explanation

Projection risk refers to the risk of misestimating future cash flows. Given the lower availability of information from private firms, the uncertainty of projected cash flows may increase.

However, management may not be experienced with projections and may underestimate *or* overestimate future prospects. The discount rate would then be decreased *or* increased accordingly. So management is not always overly optimistic and projection risk does not always result in higher discount rates.

(Module 24.2, LOS 24.e)

Question #6 of 49

Question ID: 1473333

Using the following information, calculate the WACC using the build-up method, assuming the firm is being acquired.

Income return on bonds	6.0%
Capital return on bonds	2.0%
Long-term Treasury yield	3.5%
Beta	1.4
Equity risk premium	6.0%
Small stock premium	4.0%
Company-specific risk premium	3.0%
Industry risk-premium	2.0%
Pretax cost of debt	11.0%
Optimal Debt/Total Cap	20%
Current Debt/Total	7%
Debt/Total Cap for public firms in industry	33%
Tax Rate	30%

A) 18.5%.

B) 16.3%.

C) 17.7%.

Explanation

Using the build-up method: the risk-free rate, the equity risk premium, the small stock premium, a company-specific risk premium, and an industry risk premium are added together: 3.5% + 6.0% + 4.0% + 3.0% + 2.0% = 18.5%. Note that the risk-free rate is the Treasury yield, not the returns for bonds in general.

Because the firm is being acquired, we assume the new owners will utilize an optimal capital structure and weights in the WACC calculation. The capital structure for public firms should not be used because public firms have better access to debt financing.

The WACC using the optimal capital structure factors in the debt to total cap, the cost of debt, the tax rate, and the given cost of equity:

$$[20\% \times 11\% \times (1\text{-}30\%)] + [(1\text{-}20\%) \times 18.5\%] = 16.3\%.$$

(Module 24.2, LOS 24.f)

Question #7 of 49

Question ID: 1473346

An analyst calculates a control premium of 15% and discount for lack of marketability (DLOM) of 20%. Which of the following is *closest* to the total discount for valuing minority equity interests in the private firm?

A) 35.7%.

B) 35.0%.

C) 30.4%.

Explanation

The discount for lack of control (DLOC) can be backed out of the control premium.

$$\text{DLOC} = 1 - \left[\frac{1}{1 + \text{Control Premium}}\right]$$

$$\text{DLOC} = 1 - \left[\frac{1}{1 + 0.15}\right] = 13.04\%$$

The total discount also uses the DLOM.

Total Discount = 1 − [(1 − DLOC)(1 − DLOM)]

Total Discount = 1 − [(1 − 0.1304)(1 − 0.20)] = 30.4%

(Module 24.4, LOS 24.j)

Question #8 of 49

Question ID: 1473326

Using the following figures, calculate the value of the firm using the excess earnings method (EEM).

Working capital	$600,000
Fixed assets	$2,300,000
Normalized earnings	$340,000
Required return for working capital	5%
Required return for fixed assets	13%
Growth rate of residual income	4%
Discount rate for intangible assets	18%

A) \$3,027,111.

B) \$2,981,714.

C) \$3,073,199.

Explanation

The answer is calculated using the following steps.

Step 1: Calculate the required return for working capital and fixed assets.

Given the required returns in percent, the monetary returns are:

Working Capital: \$600,000 × 5% = \$30,000.

Fixed Assets: \$2,300,000 × 13% = \$299,000.

Step 2: Calculate the residual income.

After the monetary returns to assets are calculated, the residual income is that which is left over in the normalized earnings:

Residual Income = \$340,000 − \$30,000 − \$299,000 = \$11,000.

Step 3: Value the intangible assets.

Using the formula for a growing perpetuity, the discount rate for intangible assets, and the growth rate for residual income:

Value of Intangible Assets = (\$11,000 × 1.04) / (0.18 − 0.04) = \$81,714.

Step 4: Sum the asset values to arrive at the total firm value.

Firm Value = \$600,000 + \$2,300,000 + \$81,714 = \$2,981,714.

(Module 24.2, LOS 24.g)

Question #9 of 49

Question ID: 1473336

A private pharmaceutical firm is under consideration for acquisition where the financial buyer will pay with equity. Part of the payment to the sellers is based on FDA approval of the firm's drug. If the analyst uses a market approach and comparable data from public firms, which of the following would *most likely* result in a price-multiple that is too high? The comparable data is:

A) from transactions where the buyer used cash.

B) for transactions where the consideration was non-contingent.

C) for strategic buyers.

Explanation

In market approaches, the analyst values the subject private firm using price multiples from previous public and private transactions. A strategic buyer is one who will have synergies with the target whereas a financial buyer does not. A financial transaction typically has a smaller price premium. So in this case, the comparable price-multiple will be too high.

If the acquisition involves the acquirer's stock, the acquirer may be using overvalued shares to buy their target. Using comparables where cash is the consideration would result in lower price multiples.

Contingent consideration is payment to the sellers based on the achievement of specific goals such as FDA approval. Contingent consideration increases the risk to the seller and ceteris paribus, they would demand a higher price. Using comparables where the consideration was non-contingent would result in lower price multiples.

(Module 24.3, LOS 24.h)

Stan Bowles works for Marsh Inc. and has been tasked with the valuation of Park Limited, a small private footwear producer. Bowles prepares a valuation report on Park Limited and his report contains the following:

Comment 1:	Company-specific characteristics such as the quality and depth of management, tax considerations, and shareholders agreements that restrict liquidity mark the main differences between a private and public company.
Comment 2:	The value of a private company depends on the investor's expectations and investment requirements and could differ from one buyer to the next due to different perception of risk and future potential.
Comment 3:	To obtain an appropriate discount rate for Park, we have assumed the following: • Estimated beta of Park: 0.75 • Company specific risk premium: 1.2% • Small stock premium: 2.8% • Risk free rate: 3% • Equity risk premium: 4.5%
Comment 4:	If we are valuing Park for non-controlling equity interest, a discount for lack of control might be required.

Question #10 - 13 of 49

Question ID: 1473357

Which of the following mentioned in Comment 1 is *least* likely to be a company specific characteristic of a private company?

A) Tax consideration.

B) Quality and depth of management.

C) Shareholders agreements that restrict liquidity.

Explanation

Company specific characteristics include:

- Stage in life-cycle
- Size
- Overlap of shareholders and management
- Quality and depth of management
- Quality of financial and other information
- Less pressure from short-term investors
- Tax consideration

(Module 24.1, LOS 24.a)

Question #11 - 13 of 49

Question ID: 1473358

Comment 2 is describing:

A) fair value.

B) investment value.

C) fair market value.

Explanation

Investment value is the value of a private company that depends on the investor's expectations and investment requirements. Investment value could differ from one buyer to the next due to different perception of risk and future potential.

(Module 24.4, LOS 24.j)

Question #12 - 13 of 49

Question ID: 1473359

Calculate Park's discount rate using the build-up method and the information in Comment 3.

A) 9.375%.

B) 10.375%.

C) 11.5%.

Explanation

Using the build-up method, there is no beta adjustment on the equity risk premium. The discount rate is calculated as follows:

Risk free rate:	3%
Equity risk premium:	4.5%
Company specific risk premium:	1.2%
Small stock premium:	2.8%
Return on equity:	11.5%

(Module 24.2, LOS 24.f)

Question #13 - 13 of 49

Question ID: 1473360

Using Comment 4, which of the following method is *least* likely to require a discount for lack of control?

A) Guideline Transaction Method.

B) Capitalized Cash Flow Method.

C) Guideline Public Company Method.

Explanation

Guideline Public Company Method uses the price multiples of comparable public companies as Park Limited. This method typically assumes minority control. Guideline Transaction Method uses the price multiples of acquisitions of the entire public or private companies, thus reflecting controlling interest. Depending on the cash flows and discount rate estimated, the capitalized cash flow method can be applied towards the valuation of minority or control interest.

(Module 24.4, LOS 24.j)

Question #14 of 49

Question ID: 1473348

Assume a minority shareholder holds 10% of a private firm's equity, with the CEO holding the other 90%. Using normalized earnings, the value of the firm's equity is estimated at $20 million. The CEO refuses to sell the firm and the minority shareholder cannot sell their interest easily. A discount for lack of marketability (DLOM) of 15% will be applied. A discount for lack of control (DLOC) will also be estimated. Using reported earnings instead of normalized earnings provides an estimated firm equity value of $19 million. Which of the following is *closest* to the value of the minority shareholder's equity interest?

A) $1,700,000.

B) $1,900,000.

C) $1,615,000.

Explanation

Given these figures, the value of the minority shareholder's equity interest is:

Firm's equity value	$19,000,000
Minority interest	10%
Value of minority interest without discounts	$1,900,000
minus DLOC of 0%	0
Value of interest if marketable	$1,900,000
minus DLOM of 15%	$285,000
Value of minority interest	$1,615,000

(Module 24.4, LOS 24.j)

Question #15 of 49

Question ID: 1473335

Which of the following statements related to the market approaches to private company valuation is *most accurate*:

A) The guideline public company method (GPCM) is based on price multiples from comparable traded firms.

B) The prior transaction method (PTM) is based on price multiples from the sale of whole public and private companies.

C) The guideline transactions method (GTM) is based on historical stock sales of the actual subject company.

Explanation

The guideline public company method (GPCM) approach to private company valuation uses price multiples from traded public companies with adjustments for risk differences. The guideline transactions method (GTM) uses the price multiples from the sale of whole public and private companies, again with adjustments for risk differences. The prior transaction method (PTM) uses historical stock sales of the subject company; it works best when using recent, arm's-length data of the same motivation.

(Module 24.3, LOS 24.h)

Question #16 of 49

Question ID: 1473340

When would the asset-based approach result in a higher valuation than its going concern value, in the case of private company valuation?

A) When valuing biotech firms. ✗

B) If the firm has minimal profits and poor prospects. ✓

C) When valuing pharmaceutical firms. ✗

Explanation

If a firm has minimal profits and little hope for better prospects; it might be valued more highly for its liquidation value than as a going concern if another firm can put the assets to better use. Because the asset-based approach values firm equity as the fair value of its assets minus the fair value of its liabilities, it would capture this liquidation value.

Pharmaceutical and biotech firms have a high degree of intangible assets. In these cases, the going concern value is likely to be higher than the value from the asset-based approach.

(Module 24.3, LOS 24.i)

Question #17 of 49

Question ID: 1473321

Which of the following *best* describes the use of FCFF and FCFE when used in private firm valuation?

A) FCFE is usually favored if the firm is going to change its capital structure because the cost of equity is less sensitive to leverage changes than the WACC. ✗

B) FCFF is usually favored if the firm is going to change its capital structure because the WACC is less sensitive to leverage changes than the cost of equity. ✓

C) FCFE is usually favored if the firm is going to change its capital structure because the equityholders are usually the investors requesting the valuation. ✗

Explanation

Free cash flow to the firm (FCFF) can be used to value the firm as a whole and free cash flow to equity (FCFE) can be used for equity. FCFF is usually favored if the firm is going to significantly change its capital structure. The reason is that the discount rate used for FCFF valuation, the weighted average cost of capital (WACC), is less sensitive to leverage changes than the discount rate used for FCFE valuation, the cost of equity. Thus, the FCFF valuation will not vary as much as the FCFE valuation.

(Module 24.2, LOS 24.c)

Question #18 of 49

Question ID: 1473320

An analyst is valuing a small private firm that is still developing and has yet to generate any earnings. Which of the following *best* describes the approach that should be used?

A) A market approach based on public comparables would be utilized.

B) Nonoperating assets are not crucial to the firm and should be excluded in any valuation.

C) An asset-based approach would be used.

Explanation

The valuation approach used will depend on the firm's operations and its lifecycle stage. Early in its life, a firm's future cash flows may be so uncertain that an asset-based approach would be selected. The price multiples from large public firms should not be used for a small private firm when using the market approach. Although a firm's nonoperating assets are not crucial to the firm, they should be included in any valuation.

(Module 24.1, LOS 24.d)

Question #19 of 49

Question ID: 1473347

Which of the following *best* describes the estimation of discounts for lack of marketability (DLOM) in private company valuations? The primary advantage of using put prices to estimate the DLOM over the other two methods is:

A) the Black-Scholes model has been shown to be valid for private firms.

B) exchange traded put prices are readily available.

C) the volatility of the firm can be incorporated into the analysis.

Explanation

If an interest in a firm cannot be easily sold, a DLOM is applied. The DLOM can be estimated using restricted share versus publicly traded share prices, pre-IPO versus post-IPO prices, and put prices. The advantage of using put prices over the other two DLOM estimation methods is that the estimated risk of the firm can be factored into the option price.

(Module 24.4, LOS 24.j)

Question #20 of 49

Question ID: 1473317

An analyst is valuing a firm's equity using the price-to-book-value ratio of similar firms. Which of the following is the *most likely* valuation approach the analyst will use?

A) The asset-based approach.

B) The income approach.

C) The market approach.

Explanation

The market approach values a firm using the price-multiples such as the price-to-book-value ratio and price-earnings ratio of comparable assets. The income approach values a firm as the present value of its future income. The asset-based approach values a firm as its assets minus liabilities.

(Module 24.1, LOS 24.d)

Question #21 of 49

Question ID: 1473328

Which of the following *best* describes the use of size premiums when estimating the discount rate for private company valuations?

A) The treatment is similar to that for public firms.

B) When using data from comparable public firms, a distress premium may be inadvertently added in.

C) A size premium is subtracted when calculating the discount rate.

Explanation

For private company valuations, a size premium is often added in when calculating the discount rate. This is not typically done for public firms. To get the size premium, the appraiser may use data from the smallest cap segment of public equity. This however may include a distress premium that is not applicable to the private firm.

(Module 24.2, LOS 24.e)

Question #22 of 49

Question ID: 1473314

Which of the following is *least likely* an example of a litigation-related valuation for a private company?

A) Bankruptcy proceeding.

B) Divorce settlements.

C) Lost profits claims.

Explanation

Litigation-related valuations may be required for shareholder suits, damage claims, lost profits claims, or divorce settlements. A bankruptcy proceeding is an example of a transaction-related valuation for a private company.

(Module 24.1, LOS 24.b)

Question #23 of 49

Question ID: 1473323

Given the following figures, calculate the FCFF. Assume the earnings and expenses are normalized and that capital expenditures will cover depreciation plus 3 percent of the firm's incremental revenues.

Current Revenues	$30,000,000
Revenue growth	6%
Gross profit margin	20%
Depreciation expense as a percent of sales	1%
Working capital as a percent of sales	15%
SG&A expenses	$3,800,000
Tax rate	30%

A) $1,245,400.

B) $1,785,400.

C) $927,400.

Explanation

The answer is calculated as follows:

Pro forma Income Statement	
Revenues	$31,800,000
Cost of Goods Sold	$25,440,000
Gross Profit	$6,360,000
SG&A Expenses	$3,800,000
Pro forma EBITDA	$2,560,000
Depreciation and amortization	$318,000
Pro forma EBIT	$2,242,000
Pro forma taxes on EBIT	$672,600
Operating income after tax	$1,569,400
Adjustments to obtain FCFF	
Plus: Depreciation and amortization	$318,000
Minus: Capital expenditures	$372,000
Minus: Increase in working capital	$270,000
FCFF	$1,245,400

The following provides a line by line explanation for the above calculations.

Pro forma Income Statement	Explanation
Revenues	Current revenues times the growth rate: $30,000,000 × (1.06)
Cost of Goods Sold	Revenues times one minus the gross profit margin: $31,800,000 × (1 − 0.20)
Gross Profit	Revenues times the gross profit margin: $31,800,000 × 0.20
SG&A Expenses	Given in the question
Pro forma EBITDA	Gross Profit minus SG&A expenses: $6,360,000 − $3,800,000
Depreciation and amortization	Revenues times the given depreciation expense: $31,800,000 × 0.01
Pro forma EBIT	EBITDA minus depreciation and amortization: $2,560,000 − $318,000
Pro forma taxes on EBIT	EBIT times tax rate: $2,242,000 × 0.30
Operating income after tax	EBIT minus taxes: $2,242,000 − $672,600

Adjustments to obtain FCFF	
Plus: Depreciation and amort.	Add back noncash charges from above
Minus: Capital expenditures	Expenditures cover depreciation and increase with revenues: \$318,000 + (0.03 × \$31,800,000 − \$30,000,000)
Minus: Increase in working capital	The working capital will increase as revenues increase: (0.15 × \$31,800,000 − \$30,000,000)
FCFF	Operating income net of the adjustments above

(Module 24.2, LOS 24.c)

Question #24 of 49

Question ID: 1473313

Which of the following is *least likely* an example of a transaction-related valuation for a private company?

A) Bankruptcy proceeding.

B) Financial reporting.

C) Performance-based managerial compensation.

Explanation

Venture capital financing, initial public offering (IPO), bankruptcy proceeding, performance-based managerial compensation, and sale in an acquisition are all examples of transaction-related valuations for a private company.

(Module 24.1, LOS 24.b)

Jimmy Choo, CFA, works for Joel Constable, a high net worth investor who invests a large proportion of his portfolio speculatively in order to chase high returns. Choo is usually tasked with carrying out initial analysis on investment opportunities, which Constable has identified and may proceed further with.

Most of the opportunities identified by Constable are investments in private companies with little history of profits or dividends. Some, however, are more stable growth companies, which Constable still believes to be undervalued.

One opportunity recently identified by Constable is a private company, Bakan Fashion. Bakan is a fashion retailer, which started up recently with an online presence and a retail unit in a high-end location on a city high street. The company began trading four years ago and has published the following financial information for the benefit of its investors:

Bakan Fashion

Income Statement ($m) (extracts)	**2x07**	**2x08**	**2x09**	**2x10**
Revenue	0.3	0.8	1.9	1.9
Net income	(0.9)	(0.9)	(0.3)	0.1
Cash flow statement ($m) (extracts)	**2x07**	**2x08**	**2x09**	**2x10**
CFO	(0.6)	(0.7)	(0.3)	0.2
CFI	(3.6)	(3.2)	(1.0)	(0.4)
CFF	5.6	1.0	0.4	0.4
Balance sheet ($m) (extracts)	**2x07**	**2x08**	**2x09**	**2x10**
Net assets	3.2	3.0	2.6	2.9

Choo has also made some additional notes regarding the 2x10 net income figure after meeting briefly with the finance director of Bakan.

Notes Re: 2x10 Net Income

- CEO compensation package is $650,000 per annum
- A leading fashion supplier provided the retail outlet at a rate of $120,000 per annum for the first four years
- A friend of the CEO helped set up the online operations for a discounted cost of $20,000
- The CEO agreed to the compensation package in a bid to help the company breakeven as quickly as possible. Choo estimates that a market based compensation figure would be $850,000
- The market lease rate for the retail unit is $170,000
- The set up cost for the online operation should have been $50,000

In addition to the financial information for Bakan, Choo has also identified the following specific factors, which he always takes into account in any analysis of private companies for Constable:

Factor 1:	Private company equity typically has fewer potential owners and is less liquid than publicly traded shares; hence a discount should be applied in any valuation of Bakan.
Factor 2:	One key objective of the board of Bakan may be to limit exposure to taxes, as 75% of the board members also own substantial equity in the company.
Factor 3:	Due to the significant ownership of equity by the board of Bakan, management may be tempted to take a shorter-term view than that typically taken by public firms.

Constable always prefers to see a basic analysis of price multiples for companies before any valuation.

Choo produces this where possible, but usually follows it with a valuation of the private company-using price multiples based on recent sales of comparable assets. He calls this the "Choo Comparison Method (CCM)."

The CCM valuation is then adjusted by Constable and Choo to take account of Constable's individual financing costs and any perceived synergies with his existing assets.

Choo refers to this final valuation after adjustments in his reports as the "Intrinsic Value." Constable has recently taken him to task on this and suggested that as the valuation focuses on the value to him as a specific investor it should be called the "Investment Value."

Question #25 - 30 of 49

Question ID: 1473350

Assuming Choo uses price multiples based on the four years of financial information from Bakan, ignoring his additional notes regarding 2x10, which of the following statements is least accurate?

A) Price to Earnings ratios would be meaningless.

B) Price to Book ratios would be the most useful as they would take account of the human capital in a start-up business such as Bakan.

C) Price to Cash Flow ratios are theoretically better if calculated using free cash flow to equity rather than CFO.

Explanation

As Bakan has negative earnings over the period, a P/E ratio would be negative and hence meaningless, so A is correct. Book values do not take account of human capital, as it is not on the Balance Sheet within Net Assets. FCFE is preferable to CFO.

(Module 22.1, LOS 22.c)

Question #26 - 30 of 49

Question ID: 1473351

When valuing private companies, how would the factors described by Choo usually be classified?

A) Factor 1 Stock-specific, Factor 2 Stock-specific.

B) Factor 1 Stock-specific, Factor 2 Company-specific.

C) Factor 1 Company-specific, Factor 2 Company-specific.

Explanation

Stock-specific factors are **liquidity**, restrictions on marketability, concentration of control.

Company-specific are stage of lifecycle, size, quality and depth of management, management shareholder overlap, short-term investors, quality of information, **taxes**.

(Module 24.1, LOS 24.a)

Question #27 - 30 of 49

Question ID: 1473352

Is Choo correct in his assessment of Factor 3?

A) Yes.

B) No, it is rare for private companies to allow key management to own equity.

C) No, board equity ownership usually results in a longer-term view than public companies.

Explanation

Shareholders in public firms often focus on short-term measures of performance due to pressure from institutional investors, whereas the equity ownership leads to a longer-term view for owner managers.

(Module 24.1, LOS 24.a)

Question #28 - 30 of 49

Question ID: 1473353

The "Choo Comparison Method" of valuing a private company is consistent with which major approach?

A) Income Approach.

B) Multiple Approach.

C) Market Approach.

Explanation

The three major approaches are income—pv of future incomes, market—as described by Choo and asset-based—net asset valuation.

(Module 24.1, LOS 24.d)

Question #29 - 30 of 49

Question ID: 1473354

Is Constable right in his suggested renaming of the final valuation?

A) Yes.

B) No, a valuation based on analysis of company fundamentals should be referred to as an Intrinsic Value.

C) No, it should be renamed Market Value as it uses primarily market information.

Explanation

Investment value focuses on the value to a particular buyer and is important in private company valuation.

(Module 24.4, LOS 24.j)

Question #30 - 30 of 49

Question ID: 1473355

Using Choo's additional notes on Bakan's 2x10 Net Income, which of the following is the most accurate estimate of normalized earnings for 2x10?

A) Loss $150,000.

B) Profit $250,000.

C) Loss $180,000.

Explanation

Net Income $100,000 – $200,000 – $50,000 = –$150,000

Deduct $200,000 for extra CEO compensation

Deduct $50,000 for extra lease cost

One off online set up cost is sunk and not relevant to future earnings.

(Module 24.2, LOS 24.c)

Question #31 of 49

Question ID: 1473312

Which of the following is *least likely* an example of a compliance-related valuation for a private company?

A) Tax purposes.

B) Bankruptcy proceeding.

C) Financial reporting.

Explanation

A bankruptcy proceeding is an example of a transaction-related valuation for a private company.

(Module 24.1, LOS 24.b)

Question #32 of 49

Question ID: 1473332

Using the following information, calculate the required return on equity using the expanded CAPM.

Income return on bonds	6.0%
Capital return on bonds	2.0%
Long-term Treasury yield	3.5%
Beta	1.4
Equity risk premium	6.0%
Small stock premium	4.0%
Company-specific risk premium	3.0%
Industry risk-premium	2.0%
Pretax cost of debt	11.0%
Optimal Debt/Total Cap	16%
Current Debt/Total	7%
Debt/Total Cap for public firms in industry	33%
Tax Rate	30%

A) 11.9%.

B) 18.9%.

C) 15.9%.

Explanation

The required return on equity using the CAPM is: 3.5% + 1.4(6%) = 11.9%.

Note that the risk-free rate is the Treasury yield, not the returns for bonds in general.

Using the expanded CAPM, a small stock premium and company-specific risk premium are added: 11.9% + 4% + 3% = 18.9%.

(Module 24.2, LOS 24.f)

Question #33 of 49

Question ID: 1473309

Which of the following statements *most* accurately describes the difference between private and public firm managers?

A) Because managers in a public firm are often paid with incentive compensation, public managers may take a longer term view than private managers.

B) Because managers in a private firm are concerned with having the firm go public, private managers may take a shorter term view than public managers.

C) Although managers in a public firm are often paid with incentive compensation, public managers may take a shorter term view than private managers because shareholders often focus on the short-term.

Explanation

Although managers in a public firm are often paid with incentive compensation such as options, shareholders often focus on short-term measures such as quarterly earnings and the consistency of such. Management may therefore take a shorter term view than they otherwise would. Private firms should be able to take a longer term view.

(Module 24.1, LOS 24.a)

Question #34 of 49

Question ID: 1473337

An analyst is valuing a private firm on the behalf of a strategic buyer and deflates the average public company multiple by 15% to account for the higher risk of the private firm. Given the following figures, calculate the value of firm equity using the guideline public company method (GPCM).

Market value of debt	$4,100,000
Normalized EBITDA	$42,800,000
Average MVIC/EBITDA multiple	8.5
Control premium from past transaction	25%

The value of the firm's equity is *closest* to:

A) $304,060,000.

B) $382,438,000.

C) $381,412,500.

Explanation

The adjustment to the MVIC/EBITDA multiple for the higher risk of the private firm is: 8.5 × (1 − 0.15) = 7.225. Given that the buyer is a strategic buyer, a control premium adjustment should be made on the value of equity.

MVIC = 7.225 × $42,800,000 = $309,230,000.

Subtracting out the debt results in the equity value (before control premium): $309,230,000 − $4,100,000 = $305,130,000.

Equity value after applying control premium = 305,130,000(1.25) = 381,412,500

(Module 24.3, LOS 24.h)

Question #35 of 49

Question ID: 1473322

Given the following figures, calculate the normalized EBITDA for a financial and strategic buyer.

Reported EBITDA	$4,500,000
Current Executive Compensation	$700,000
Market-Based Executive Compensation	$620,000
Current SG&A expenses	$6,300,000
SG&A expenses after synergistic savings	$5,600,000
Current Lease Rate	$300,000
Market-Based Lease Rate	$390,000

The normalized EBITDA for each type of buyer is:

	Financial Buyer	Strategic Buyer
A)	$4,190,000	$4,890,000
B)	$4,490,000	$5,190,000
C)	$4,670,000	$5,370,000

Explanation

Both strategic and financial buyers will attempt to reduce executive compensation to market levels by \$80,000 (\$700,000 − \$620,000). They will also have to pay a higher lease rate of \$90,000 (\$390,000 − \$300,000). So the adjustment for both buyers to generate normalized EBITDA is \$4,500,000 + \$80,000 − \$90,000 = \$4,490,000.

However, only a strategic buyer will be able to realize synergistic savings of \$700,000 (\$6,300,000 − \$5,600,000). So normalized EBITDA for a strategic buyer is \$5,190,000 and for a financial buyer it is \$4,490,000.

(Module 24.2, LOS 24.c)

Question #36 of 49

Question ID: 1473315

Assume that a property that you are evaluating has a gross annual income equal to \$230,000, and that comparable properties are selling for 10.5 times gross income. The gross income multiplier approach provides a market value for this property that is *closest* to:

A) \$2,415,000. ✓

B) \$2,190,000 ✗

C) \$2,303,000. ✗

Explanation

Gross income multiplier technique: MV = gross income × income multiplier.

MV = \$230,000 × 10.5 = \$2,415,000

(Module 24.1, LOS 24.b)

Question #37 of 49

Question ID: 1473318

Which of the following approaches to private company valuation uses discounted cash flow analysis?

A) The asset-based approach. ✗

B) The market approach. ✗

C) The income approach. ✓

Explanation

The income approach values a firm as the present value of its future income. The asset-based approach values a firm as its assets minus liabilities. The market approach values a firm using the price-multiples from the sales of comparable assets.

(Module 24.1, LOS 24.d)

Question #38 of 49

Question ID: 1473334

An analyst values a private firm by using price multiples from the sale of whole companies, with adjustments for risk differences. Which of the following best describes the valuation method that the analyst is using?

A) The guideline transactions method.

B) The prior transaction method.

C) The guideline public company method.

Explanation

The guideline transactions method (GTM) generates a value estimate based on pricing multiples associated with the acquisition of control of entire companies. The guideline public company method (GPCM) generates an estimate of value based on the multiples from trading activity in the shares of public companies that are similar to the private company in question. The prior transaction method (PTM) uses actual transactions in the stock of the subject private company.

(Module 24.3, LOS 24.h)

Question #39 of 49

Question ID: 1473311

An analyst is examining the stock of three companies. Given the information below, which of them is *most likely* to be the stock of a private firm?

Firm	***Restrictions on Sale of Stock?***	***DLOM***	***Stock Ownership of 5 Largest Owners***
A	Yes	0%	28%
B	No	5%	35%
C	Yes	15%	64%

A) Firm A.

B) Firm B.

C) Firm C.

Explanation

The stock most likely to be that of a private firm is Firm C. Compared to public stock, private firm stock often has agreements that prevent shareholders from selling, is less liquid (discounts for lack of marketability (DLOM) of C is 15%), and control is usually concentrated in the hands of a few shareholders (stock ownership of largest owners of Firm C is 64%).

(Module 24.1, LOS 24.a)

Question #40 of 49

Question ID: 1473310

An analyst is examining three companies. Given the information below, which of them is *most likely* to be a private firm?

Firm	Number of Years in Operation	Market Capitalization	Required Return for Common Stock
A	12 years	$1,324.8 million	14.8%
B	4 years	$1,313.9 million	18.3%
C	19 years	$2,231.0 million	16.4%

A) Firm A.

B) Firm B.

C) Firm C.

Explanation

The firm most likely to be a private firm is Firm B. Compared to public firms, private firms are less mature (4 years for Firm B), smaller (market cap of B is $1,313.9 million), and have higher required returns (required return for B is 18.3%).

(Module 24.1, LOS 24.a)

Question #41 of 49

Question ID: 1473338

The asset-based approach values a firm based on:

A) investment values.

B) book values.

C) fair values.

Explanation

The asset-based approach values firm equity as the fair value of its assets minus the fair value of its liabilities.

(Module 24.3, LOS 24.i)

Paul Smith is an analyst performing valuations for Lumber Limited. Smith has been given a project to value Timber Industries, a firm that Lumber Limited is considering acquiring. Smith is aware that a number of characteristics distinguish private and public companies, and that these characteristics must be considered during his process of valuing Timber Industries. A number of issues complicate Smith's valuation: Timber Industries pays its CEO well below a market-based compensation figure, leases a warehouse at an above-market rate, and owns a vacant office building that is not needed for core operations. Smith is also aware that discounts and premiums based on control and marketability must be considered in his valuation of Timber Industries.

Question #42 - 45 of 49

Question ID: 1473342

Compared to a public company, it is *most likely* that as a private company Timber Industries will have greater:

A) focus on the short-term.

B) quality and depth of management.

C) concerns related to taxes.

Explanation

Private firms may be more concerned with taxes than public firms due to the impact of taxes on private equity owners/managers. Private firms are likely to have lower quality and depth of management, as private firms are likely to be smaller and thus may not be able to attract as many qualified applicants as public firms. Private firms are more likely to focus on the long-term than public companies, since in most private firms, external shareholders have less influence and the firm is able to take a longer-term perspective.

(Module 24.4, LOS 24.j)

Question #43 - 45 of 49

Question ID: 1473343

One valuation method that Smith is considering for Timber Industries involves using a growing perpetuity formula to estimate the value of intangible assets, and then adding this value to the values of working capital and fixed assets. This method is *most accurately* described as the:

A) free cash flow method. ✗

B) excess earnings method. ✓

C) capitalized cash flow method. ✗

Explanation

The excess earnings method values tangible and intangible assets separately; this method is useful for small firms and when there are intangible assets to value. In the free cash flow method, a firm is valued by discounting a series of discrete cash flows plus a terminal value. In the capitalized cash flow method, a firm is valued by discounting a single cash flow by the capitalization rate.

(Module 24.4, LOS 24.j)

Question #44 - 45 of 49

Question ID: 1473344

The asset-based approach to private company valuation that Smith is considering for Timber Industries is *most likely* to be appropriate in the case of a:

A) finance firm such as a bank. ✓

B) mature company with many intangible assets. ✗

C) firm with strong profits and growth potential. ✗

Explanation

The asset-based approach is usually not used for most going concerns, but is appropriate for troubled firms, finance firms, investment companies, firms with few intangible assets, and natural resource firms. It values equity as the asset value of a firm minus the debt value of the firm.

(Module 24.4, LOS 24.j)

Question #45 - 45 of 49

Question ID: 1473345

Which of the following statements related to discounts and premiums to benchmark for Smith's private company valuation of Timber Industries is *most accurate*?

A) A discount for lack of marketability should be applied when the comparables are based on public shares, and the interest in the target company is a minority interest in a private firm.

B) A discount for lack of control should be applied when the comparable company values are for public shares, and the target company valuation is for a controlling interest.

C) A control premium should be added when the comparable values are for the sale of an entire company, and the valuation is being done for a minority interest in the target company.

Explanation

Discounts for lack of marketability are applied when the comparables are based on highly marketable securities, such as public shares, and the interest in the target company is less marketable, as in the case of a minority interest in a private firm. A discount for lack of control is applied when the comparable values are for the sale of an entire company, and the valuation is being done for a minority interest in the target company. A control premium is added when the comparable company values are for public shares or other minority interests, and the target company valuation is for a controlling interest.

(Module 24.4, LOS 24.j)

Question #46 of 49

Question ID: 1473325

Using the following figures, calculate the value of the equity using the capitalized cash flow method (CCM), assuming the firm will be acquired.

Normalized FCFE in current year	\$3,000,000
Reported FCFE in current year	\$2,400,000
Growth rate of FCFE	7.0%
Equity discount rate	16.0%
WACC	13.0%
Risk-free rate	3.5%
Cost of debt	10.5%
Market value of debt	\$3,000,000

The value of the equity is:

A) \$35,666,667.

B) \$32,666,667.

C) \$28,533,333.

Explanation

To arrive at the value of the equity using the CCM, it can be estimated using the free cash flows to equity and the required return on equity (r):

$$\text{value of equity} = \frac{\text{FCFE}_1}{r - g}$$

$$\text{value of equity} = \frac{\$3{,}000{,}000 \times (1.07)}{0.16 - 0.07} = \$35{,}666{,}667$$

Note that we grow the FCFE at the growth rate because the *current* year FCFE is provided in the problem (not next year's FCFE). We use normalized earnings, not reported earnings, given that normalized earnings are most relevant for the acquirers of the firm. The relevant required return for FCFE is the equity discount rate, not the WACC.

An alternative approach to calculate the value of the equity would be to subtract the market value of the firm's debt from total firm value. However, the FCFF are not provided so a total firm value cannot be calculated.

(Module 24.2, LOS 24.g)

Question #47 of 49

Question ID: 1473324

The capitalized cash flow method (CCM) used in private firm valuation is *most* appropriate when:

A) earnings are growing quickly in an initial period.

B) there are many intangible assets to value.

C) stable growth is expected.

Explanation

The CCM is a growing perpetuity model that assumes stable growth and is in effect a single-stage free cash flow model. It may be suitable when no comparables or projections are available and when stable growth is expected. The excess earnings method (EEM) is useful when there are intangible assets to value. The free cash flow method assumes high growth in an initial period followed by constant growth thereafter.

(Module 24.2, LOS 24.g)

Question #48 of 49

Question ID: 1473331

Which of the following statements related to the models used to estimate the required rate of return to private company equity is *most accurate*:

A) The build-up method begins with betas for comparable public firms and adds risk premiums.

B) The CAPM model uses betas estimated from firm returns of other private firms.

C) The expanded CAPM model adds premiums for size and firm-specific risk.

Explanation

Expanded CAPM adds premiums for size and firm-specific risk. CAPM may not be appropriate for private firms because beta is usually estimated from public firm returns. The build-up method adds an industry risk and other risk premiums to market rate of return; it is used when betas for comparable public firms are not available.

(Module 24.2, LOS 24.f)

Question #49 of 49

Question ID: 1473339

Which of the following is *most* accurate regarding the asset-based approach? Of the three valuation methods for private firms, it usually:

A) is the most appropriate for going concerns.

B) results in the lowest valuation.

C) is not difficult to apply.

Explanation

The asset-based approach is generally not used for going concerns. Because it is easier to find comparable data at the firm level compared to the asset level, the income and market approaches would be preferred to value going concerns.

Because it is difficult to find data for individual intangible assets and specialized assets, the asset-based approach can be difficult to apply. It generally results in the lowest valuation because the use of a firm's assets in combination usually results in greater value creation than each of its parts individually.

(Module 24.3, LOS 24.i)